Listing 10.19 `semantic_function.py`

```
@tool
def my_python_tool(
    input: str,
    input_node: int,
    history: str,
    semantic_function: str,
    evaluation_function: str,
    function_name: str,
    skill_name: str,
    max_tokens: int,
    temperature: float,
    deployment_name: str,
    connection: Union[OpenAIConnection,
                      AzureOpenAIConnection],     #1
) -> str:
    if input is None or input == "":     #2
        return ""

    kernel = sk.Kernel(log=sk.NullLogger())
    # code for setting up the kernel and LLM connection omitted


    function = kernel.create_semantic_function(
                              semantic_function,
                              function_name=function_name,
                              skill_name=skill_name,
                              max_tokens=max_tokens,
                              temperature=temperature,
                              top_p=0.5)     #3
    evaluation = kernel.create_semantic_function(
                              evaluation_function,
                              function_name="Evaluation",
                              skill_name=skill_name,
                              max_tokens=max_tokens,
                              temperature=temperature,
                              top_p=0.5)     #4

    async def main():
        query = f"{history}\n{input}"
        try:
            eval = int((await evaluation.invoke_async(query)).result)
            if eval > 25:     #5
                return await function.invoke_async(query)    #6
        except Exception as e:
            raise Exception("Evaluation failed", e)

       try:
        result = asyncio.run(main()).result
        return result
    except Exception as e:
        print(e)
        return ""
```

#1 Uses a union to allow for different types of LLM connections
#2 Checks to see if the input is empty or None; if so, the function shouldn't be executed.
#3 Sets up the generation function that creates a plan
#4 Sets up the evaluation function
#5 Runs the evaluate function and determines if the input is good enough to continue
#6 If the evaluation score is high enough, generates the next step